

Wind Portfolio Performance Optimization

Generation Gap Recovery & Decision Support Framework

A structured analytics initiative to identify, quantify, and recover addressable energy generation deficits across a 1 GW wind portfolio. Utilizing high-frequency SCADA telemetry and machine learning models to recover 66.6 – 99.9 GWh/year with zero new capital expenditure.

1 GW

PORTFOLIO CAPACITY

38%

BASELINE CAPACITY FACTOR

2–3%

TARGET ENERGY RECOVERY

Executive Summary & Quantified Baselines

Portfolio Operational Profile

- **Total Capacity:** 1,000 MW (1 GW) distributed clean energy asset base.
- **Baseline Annual Yield:** ~3.33 TWh/year operating at a 38% capacity factor.
- **Underperforming Hours:** ~14% of operational hours exhibit a >10% power generation deficit compared to theoretical curves.

Financial & Revenue Gap

- **Generation Deficit:** 5–8% annual generation gap fleet-wide.
- **Financial Impact:** Multi-million dollar revenue loss per year due to uncaptured energy.
- **Strategic Constraint:** Zero-CapEx mandate requires optimization via software, control tuning, and targeted maintenance.

Core Strategic Objective

Develop an end-to-end analytical pipeline to isolate controllable technical losses (yaw misalignment, pitch errors, derating) from non-controllable factors (wind shear, grid curtailment), aiming to recover **66.6 to 99.9 GWh/year**.

3.33 TWh

Annual Baseline Output

5–8%

Annual Generation Loss

+99.9 GWh

Recovery Target

Build Analytical Team & Skill Triad

1. Domain Expertise

Wind Energy SMEs & Operational Leads

- Validate aerodynamic constraints & power curves.
- Define operational rules and alarm logic.
- Root-cause validation for pitch/yaw issues.

2. Data Expertise

Data Engineers & SCADA Specialists

- Ingest high-frequency 10-min SCADA telemetry.
- Schema validation & missing value imputation.
- Engineer atmospheric & spatial features.

3. Analytics Expertise

Data Scientists & MLOps Engineers

- Build time-aware regression & XGBoost models.
- Implement SHAP attribution pipelines.
- Enforce model accuracy threshold ($R^2 \geq 0.90$).

Governance & Project Management Structure

Lead Data Scientist / Project Lead: Manages CRISP-DM lifecycle, aligns analytical targets with business goals, and conducts stage exit gates.

BI & CMMS Integration Developer: Constructs fleet visualization dashboards and automates dispatch work orders for field technicians.

Understand Core Objectives

1. Power Curve Forecasting

Predict continuous *Expected_Power_kW* under real-time weather conditions and forecast short-horizon generation (1h, 6h, 24h).

2. Automated Deficit Flagging

Automatically flag periods where turbine generation falls >10% below theoretical capacity during eligible operating windows.

3. Loss Attribution & Separation

Isolate non-controllable losses (weather extremes, grid curtailment) from addressable technical faults (yaw offset, pitch error).

4. Early Anomaly Detection

Monitor multi-sensor temperature and vibration telemetry to detect component degradation 14–30 days prior to breakdown.

5. Monetized Work Prioritization

Rank underperforming turbines by financial loss impact ($Lost_MWh \times Price$) to maximize ROI on maintenance dispatch.

6. CMMS Integration

Deliver actionable decision-support tools and direct integration with Computerized Maintenance Management Systems.

Question Bank: Descriptive & Diagnostic

Framing problem statement: "Why are specific turbines producing less power than expected, and which losses are addressable?"

1. Descriptive Questions

- **Yield Baseline:** What is the historic capacity factor and yield per turbine over 12 months?
- **Deficit Ratio:** What fraction of operational hours suffered >10% underperformance?
- **Loss Distribution:** How are MWh losses distributed across farms, regions, and OEMs?
- **Environmental Context:** What weather conditions predominated during major losses?
- **Maintenance Ratio:** What is the ratio of scheduled downtime vs forced outages?

2. Diagnostic Questions

- **Anomalies:** Which pitch, yaw, or temp anomalies correlate with power gaps?
- **Attribution:** How much loss is addressable technical vs non-controllable grid/weather?
- **Control Logic:** Are software settings causing premature thermal derating?
- **Wake Effects:** Are upstream turbine wakes degrading downstream performance?
- **Curtailement:** What loss is TSO grid-mandated vs internal component protection?

Question Bank: Predictive, Prescriptive & Cognitive

3. Predictive

- What is expected fleet output over 1h, 6h, 24h?
- Which turbines risk underperformance in next 7 days?
- Can temp drift forecast failure 14-30 days out?
- What is projected quarterly MWh loss unaddressed?
- What is probability of grid curtailment?

4. Prescriptive

- Which turbines to prioritize this week for max ROI?
- What exact yaw/pitch offsets restore nominal curve?
- How to schedule repair in low-wind forecast windows?
- How to balance re-tuning vs physical overhaul?
- What automated threshold should trigger CMMS ticket?

5. Cognitive

- How to dynamically adapt power curves for turbine aging?
- Can AI separate blade icing from physical damage?
- How to ingest field repair logs to refine accuracy?
- How to trade off short-term revenue vs asset lifespan?
- How to convert telemetry into plain-text repair steps?

Define Success Measurements

Business Success Criteria

- **Energy Recovery Target:** Recapture 2–3% **Annual Energy Production** across portfolio (66.6 – 99.9 GWh/year).
- **Financial Return:** Zero new CapEx; full ROI driven by software optimization and improved O&M prioritization.
- **Operational Efficiency:** Reduction in unplanned downtime and derated operating hours.

Analytical & Technical Criteria

- **Regression Accuracy:** Power baseline model achieves $R^2 \geq 0.90$ on test validation split.
- **Detection Precision:** High precision on identifying valid >10% underperformance events without false alarms.
- **Model Governance:** Full SHAP explainability and zero data leakage between predictors and targets.

Quality & Operational Standards

Model performance must be verified across all seasons and terrain types. All recommendations must pass safety and mechanical limits established by site engineers prior to deployment.

Unit of Analysis & Target Variable Math

Atomic Observation Grain

1 Row = 1 Physical Wind Turbine observed at a 10-Minute SCADA Timestamp

Spatial Aggregation: Turbine → Site Zone → Wind Farm (*Farm_ID*) → Portfolio

1. Primary Regression Target

Expected_Power_kW (Continuous Float)

Theoretical power output predicted under real-time wind speed, air density, ambient temperature, and turbulence conditions.

2. Secondary Classification Target

Underperformance_Flag (Binary 0/1)

Identifies operating periods suffering from significant addressable generation deficits.

Mathematical Definition: Underperformance_Flag

```
Underperformance_Flag = 1 IF [  
  (Expected_Power_kW - Actual_Power_kW) / Expected_Power_kW > 0.10  
  AND Availability_Flag == True  
  AND Curtailment_Flag == False  
  AND Wind_Speed_ms >= 3.0 m/s  
] ELSE 0
```

5-Stage Machine Learning Pipeline

Stage 1: Expected Power Regression — Time-aware XGBoost/LightGBM model predicting baseline power curve under clean operational states.

Stage 2: Underperformance Classification — Detects >10% generation gaps during eligible, non-curtailed operating windows.

Stage 3: SHAP Loss Attribution — Disambiguates non-controllable environmental factors (60%) from addressable technical faults (40%).

Stage 4: Unsupervised Anomaly Detection — Multi-sensor health tracking (vibration, thermal drift) to capture early mechanical wear.

Stage 5: Prescriptive Ranking & Monetization — Ranks turbines by financial loss impact ($Lost_MWh \times Price$) to guide maintenance dispatch.

Non-Controllable Losses (~60%)

Weather Extremes (35%), Grid Curtailment (25%).

Controllable Losses (~40%)

Yaw Misalignment (18%), Pitch Drift (14%),
Component Wear (8%).

Analytical Base Table (ABT) Schema

Domain Source	Sample Variables	System	Role & Description
Asset Identity	Record_ID, Turbine_ID, Farm_ID	SCADA	Primary keys and composite index.
Asset Metadata	Turbine_Model, Lat, Long, Hub_Height	GIS / Asset	OEM specs and spatial wake context.
Meteorological	Wind_Speed_ms, Wind_Direction_deg	SCADA	Primary power curve inputs.
Atmospheric	Ambient_Temp_C, Air_Density_kgm3	Met Mast	Density and seasonal corrections.
Aerodynamic	Nacelle_Pos_deg, Yaw_Offset_deg	Derived	Alignment relative to wind vector.
Control Telemetry	Pitch_Angle_deg, Rotor_RPM, Gen_RPM	Controller	Mechanical control tracking features.
Thermal Health	Gearbox_Temp_C, Gen_Temp_C	Sensors	Subsystem health and thermal derating.
Operational State	Availability_Flag, Curtailment_Flag	SCADA	Filtering uncurtailed operating hours.
Targets & ROI	Expected_Power_kW, Deficit_Flag	Model/Target	Primary regression & classification targets.

4-Phase Deployment Roadmap

Phase 1: Offline PoC & Validation

Timeline: Months 1–2

- Train regression models on historic ABT.
- Verify accuracy threshold ($R^2 \geq 0.90$).
- Conduct historical backtesting on past outages.

Phase 2: Pilot Deployment & Shadow Mode

Timeline: Months 3–4

- Batch daily scoring across 2 pilot farms (200 MW).
- Shadow mode scoring alongside site teams.
- Field verification of top flagged yaw/pitch issues.

Phase 3: Dashboard & CMMS Integration

Timeline: Months 5–6

- Stream 10-min SCADA to live monitoring dashboard.
- Automate work order creation in CMMS.
- Train regional site teams on decision support tools.

Phase 4: Full Fleet Rollout & MLOps

Timeline: Months 7+

- Expand across entire 1 GW portfolio.
- Implement continuous drift detection pipelines.
- Feedback loops from technician work order logs.

STAGE EXIT GATE

CRISP-DM Stage 1 Approval Sign-Off

Exit Gate Item	Status	Verification Details
1. Problem Statement	✓ Complete	Cause-neutral problem framing agreed.
2. Project Team	✓ Complete	Domain, Data, and Analytics leads assigned.
3. Business Objectives	✓ Complete	6 core objectives & 66.6+ GWh target set.
4. 5x5 Question Bank	✓ Complete	25 questions established across 5 families.
5. Unit of Analysis & Target	✓ Complete	10-min SCADA grain & math target defined.
6. ABT Blueprint	✓ Complete	28-feature schema matrix approved.

APPROVAL: READY TO PROCEED TO CRISP-DM STAGE 2 (DATA UNDERSTANDING)